

Algorithmic Pseudocode Sample 51

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 2 Compute expected utility

Input: $\mathbf{c}, s_0, t_0, \alpha, \beta, a, M, \rho, mcost, icost$

```
1: Compute  $tmcost$  and  $ticost$ 
2:  $ancost = tmcost + ticost$ 
3:  $util = 0$ 
4: for  $i \in \{1, \dots, M\}$  do
5:   if  $(c(i) = 1)$  OR  $(c(i) = 2)$  then
6:      $s_0 = s_0 \times \exp(-\alpha(i))$ 
7:      $t_0 = t_0 + \beta(i)$ 
8:   end if
9: end for
10:  $s = 1 - s_0$ 
11:  $t = t_0$ 
12: for  $i = 1, \dots, M$  do
13:   Generate  $u \sim U(0, 1)$ 
14:   if  $u < s$  then
15:      $cost(i) = 0$ 
16:   else
17:      $cost(i) \sim \text{Gamma}(a, t)$ 
18:   end if
19:    $cost(i) = cost(i) + ancost$ 
20:    $util = util + (1 - \exp(\rho \times cost(i)))$ 
21: end for
22: Return:  $util/M$ 
```
